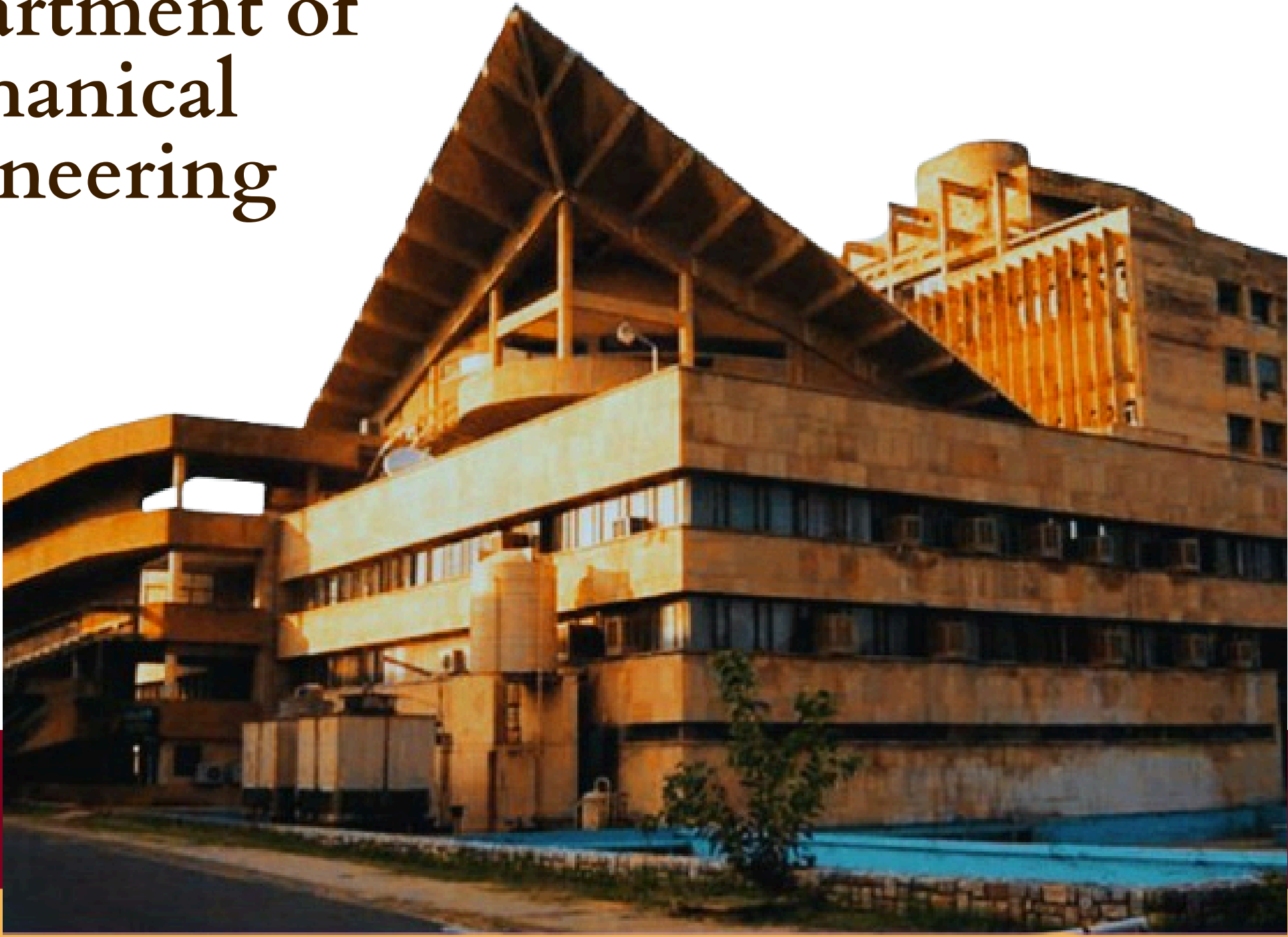




Recruitment Brochure

Department of Mechanical Engineering

IIT Delhi





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Prof. SUBBARAO P M V

Head of Department, Department of Mechanical Engineering



Department of Mechanical Engineering at IIT Delhi, an Institution of Eminence, is well known for its pursuit of excellence, reflected by its consistent upward trajectory in global standings—in the 2026 QS World University Rankings by Subject, it is ranked 44th in the world. Students of the Department strive for excellence in academics as well as extracurricular activities. Our alumni make us proud with outstanding achievements in their pursuits in academia, industry, and entrepreneurial ventures. On behalf of the Department, I welcome you to explore our talent pool of UG, PG, Ph.D., and VLFM students, and I am sure you would be delighted with their capabilities.

The department has four areas of specialization: **Mechanical Design, Production Engineering, Thermofluids** and **Industrial Engineering**. We engage in interdisciplinary research spanning a broad spectrum of domains, integrating technology, data-driven approaches, and system-level thinking to address real-world challenges across science, engineering, and societal applications. We also have many established CoEs that collaborate with industries to solve many research problems. All these facilities provide the students with ample opportunities to take part in research and be an asset for the industry.



About IIT Delhi

The Indian Institute of Technology Delhi stands as one of the 23 premier IITs established as national Centres of Excellence for advanced pedagogical training, scholarly research, and technological development across science, engineering, and technology domains.

Founded as the College of Engineering & Technology in 1961, the Institute was subsequently declared an Institution of National Importance under the 'Institutes of Technology (Amendment) Act, 1963', renamed 'Indian Institute of Technology Delhi', and granted autonomous Deemed University status with full legislative powers over its academic policy, examination frameworks, and degree conferrals. The institution was further designated an Institute of Eminence (IoE) by the Government of India.

Since its inception, close to 66,000 students have graduated from IIT Delhi across diverse academic disciplines, including Engineering, Physical Sciences, Management, Design, Public Policy, and Humanities & Social Sciences. Within this historical cohort, 22,336 individuals have been conferred Bachelor of Technology (B.Tech.) degrees and 8,038 individuals have received doctoral (Ph.D.) degrees, with the remaining balance receiving Master's degrees in Engineering, Sciences, and Business Administration.

This global alumni network actively functions as leading scientists, technologists, corporate executives, and deep-tech entrepreneurs. A significant cross-section has transitioned into public policy, administrative services, governance, and non-governmental organizations (NGOs), driving socioeconomic transformation, nation-building, and industrialization both domestically and worldwide



Department of Mechanical Engineering



The Department of Mechanical Engineering at IIT Delhi is globally recognized for its academic excellence and innovative curriculum, attracting top-ranking students through IIT-JEE and GATE. With over **1,000 students**, it offers B.Tech. programs in Mechanical Engineering and Production & Industrial Engineering, and M.Tech. specializations in Thermal

Engineering, Mechanical Design, Production Engineering, and Industrial Engineering. Over **200 students** pursue M.S. (Research) and Ph.D., contributing to cutting-edge research. The Department also offers the prestigious PGDEx-VLFM program in collaboration with IIM Mumbai for industry professionals. Graduates are equipped with strong technical and research skills, making them valuable assets in both industry and academia.

<https://mech.iitd.ac.in/>

The Department enjoys an international QS Mechanical Engineering ranking of **51–75** and a broad Engineering & Technology ranking of **26 globally (QS 2025)**. Four major areas guide our work:

Thermal Engineering

CFD, Heat Transfer, HVAC, Combustion, Turbomachinery, Energy Systems

Design Engineering

FEA, Robotics, Biomechanics, Vibration, Rotor Dynamics, Tribology

Production Engineering

Additive Manufacturing, Digital Twin, IIoT, Smart Manufacturing, Materials

Industrial Engineering

OR, Analytics, Supply Chain, Optimization, ML, Public Systems

Department of Mechanical Engineering

The department is globally recognized for its academic excellence and innovative curriculum, attracting top-ranking students through IIT-JEE and GATE.

With over 1,000 students, M.Tech., M.S.(R), Ph.D. and the prestigious PGDEx-VLFM program, the Department produces graduates equipped with strong technical and research skills.



Recent Projects

- › Energy storage system (second-life battery)
- › Composite 3D Printing – VAT Photopolymerization
- › Metal additive manufacturing for EVs
- › Machining special-grade auto materials
- › Digital Twin for Process Industry
- › IIoT for connected CNC machines
- › MSME readiness for Industry 4.0
- › ML-based remaining useful lifespan prediction
- › Hybrid EV layout for Delhi NCR traffic
- › HVAC Cooling/Heating System Analysis
- › Dual-mode scramjet engine CFD
- › Digital Twin for Vibratory System

Courses We Study

- › Computational Methods & Additive Mfg.
- › Computer Aided Manufacturing
- › Metal Forming & Machining Processes
- › Mfg. Informatics & Operation Planning
- › CAD and Finite Element Analysis
- › Industrial Engineering Systems
- › Continuum Mechanics & Analytical Dynamics
- › Intro. to Machine Learning & Neural Systems

Labs We Have

- › Foundation for Smart Manufacturing (FSM) Lab
- › Industrial Engineering & OR (IEOR) Lab
- › Integrated Design & Manufacturing (IDM) Lab
- › Production Engineering Laboratory
- › Micro and Nano Fluidics Laboratory
- › Rapid Prototyping Laboratory
- › Automation Laboratory

Why Us?

- 1 Continuously updated curricula to meet modern industrial needs — blending theory with practical application and emerging technologies.
- 2 Students gain hands-on experience with state-of-the-art labs and software and are well- trained to meet complex challenges.
- 3 Exposure to Government and industry-sponsored projects gives students practical experience and enhances their analytical and critical thinking.
- 4 Students engage in clubs, large group projects and extracurricular activities to gain excellent soft skills.
- 5 Strong alumni network with a history of industry leadership — graduates hold positions at top global firms and research institutions.

Programs

● Research Programs (Ph.D. & M.S.(R))

Ph.D. and MS(R) programs in Mechanical Engineering focusing on design, ThermoFluid, production, AI, robotics, and energy, with strong industry and government-sponsored support. These programs prepare students for careers in cutting-edge research and academia.

● Master of Technology Programs

Four M.Tech. programs with strong industry-oriented curricula:

- **Mechanical Design** – modern machinery, robotics, rotor dynamics, medical devices. **Industrial Engineering** – analytics, supply chains, logistics, healthcare, finance. **Production Engineering** – Smart Manufacturing, Industry 4.0, Additive Manufacturing, Digital Twin, Precision Metrology.
- **Thermofluids Engineering** – thermodynamics, fluid mechanics, heat transfer, energy systems.

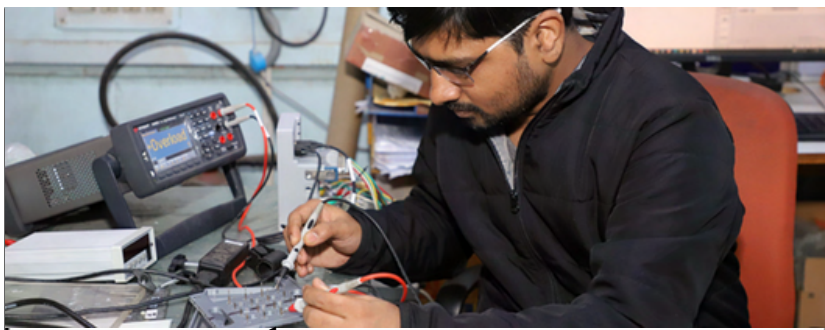
● Bachelor of Technology Programs

- **B.Tech. in Mechanical Engineering** – broad-based design, ThermoFluid, analytical skills, hands-on learning.
- **B.Tech. in Production and Industrial Engineering** – manufacturing systems, operations research, industrial engineering.

Also offered: **B.Tech. with Advanced Standing** (5-year integrated B.Tech. + M.Tech.).

● PGDEx-VLFM

A one-year, full-time executive program jointly offered by IIM Mumbai and IIT Delhi to develop visionary leaders in manufacturing. Supports Make in India and Atmanirbhar Bharat initiatives with 1300+ contact hours, industry internships, and global exposure.



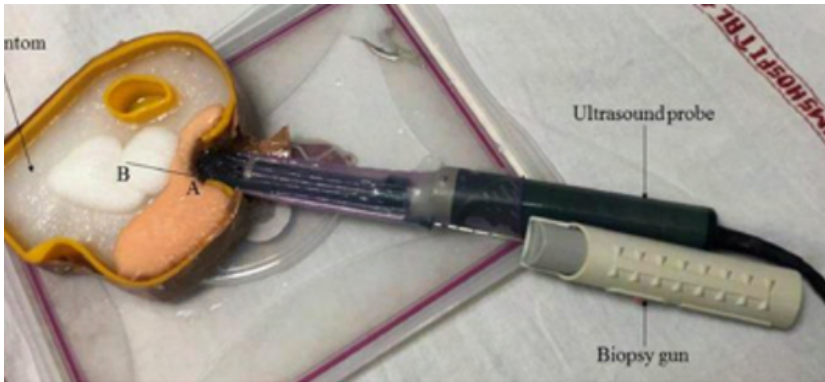
Control Systems Engineering



CT Scanner for Implants

Glimpses of our work / Research

Design Engineering



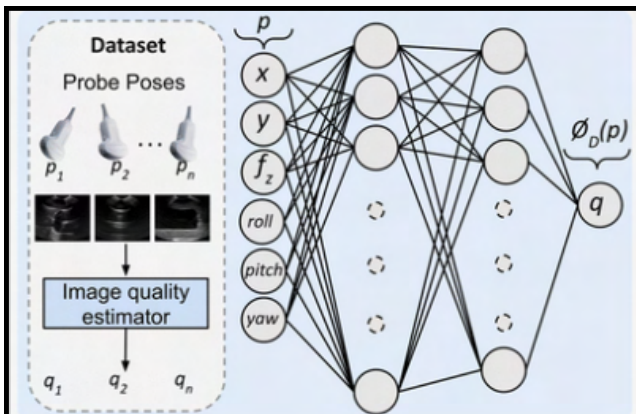
Prostate Needle Biopsy



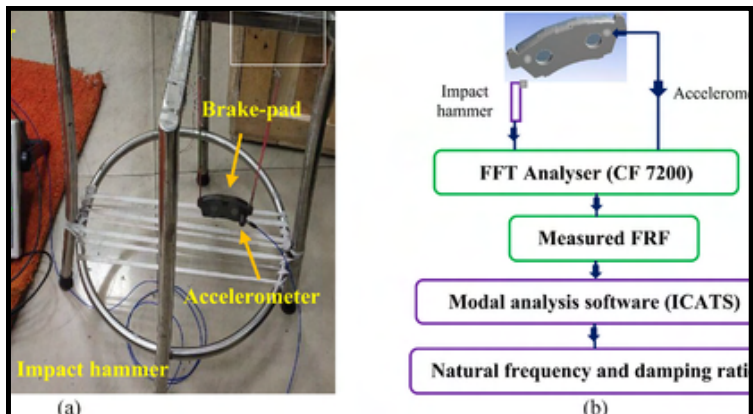
Bioengineering



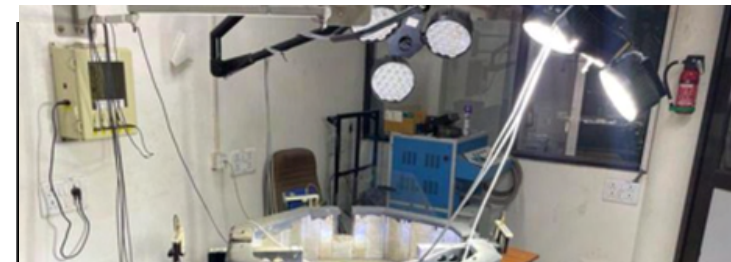
System Design



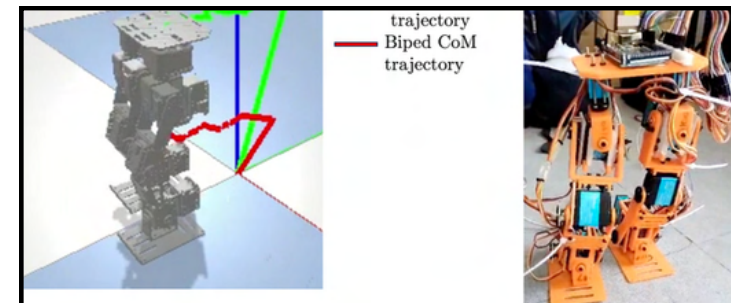
Applied Artificial Intelligence



Noise and Vibration Control for brake pads using binders



High strain rate axial test system



Robotics and Control

Mechanical Design

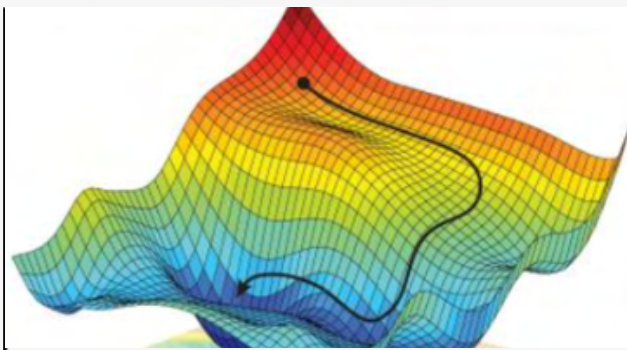
Glimpses of our Projects



- Development of a Digital Twin for a Vibratory System.
- Neural Network based System Identification.
- Computer Vision system for 3D dimension measurement and defect detection on conveyor line.
- Damage identification in composite structures using machine learning. Fracture analysis of hybrid bonding in semiconductor packaging.
- Design of a Hybrid Hydraulic Vehicle (HHV) layout for power system of urban buses.
- Development and Analysis of Quasi-Zero Stiffness (QZS) Isolators for Vibration Control
- Digital twin development for composite structures.
- Development of gravity energy storage systems.
- Modelling and experimental analysis of membrane filtration assisted by fluid vibration.
- RoboMuse 5.0: A patrolling robot.
- Studying effect of occupant personalization in car frontal crashes. Dynamics of a bolted gas turbine rotor.
- Modeling and Simulation of Environmental Control Unit of an aircraft. Design and Development of a Continuum Robot.



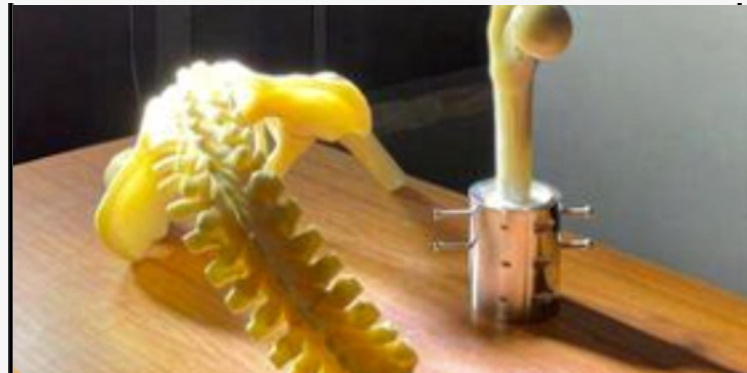
Manufacturing Technologies



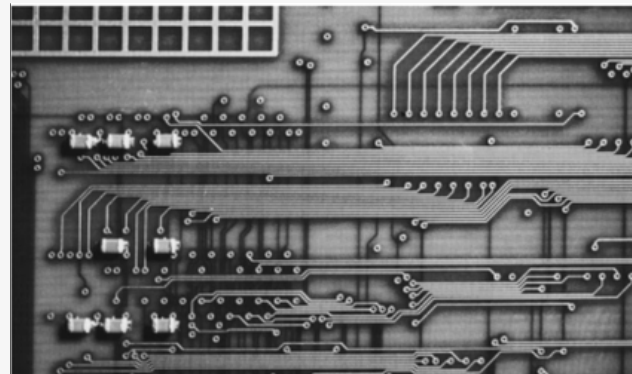
AI & Informatics



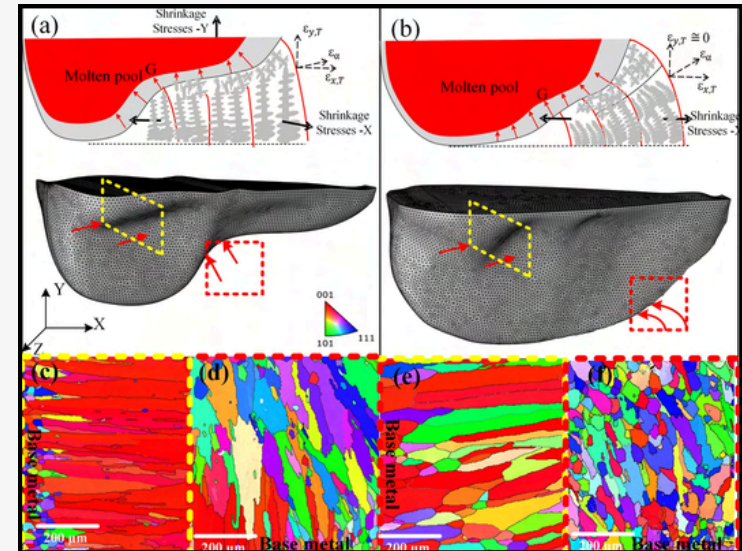
Automation and Robotics



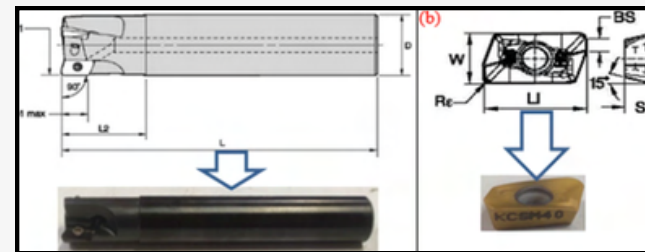
Composite Implants



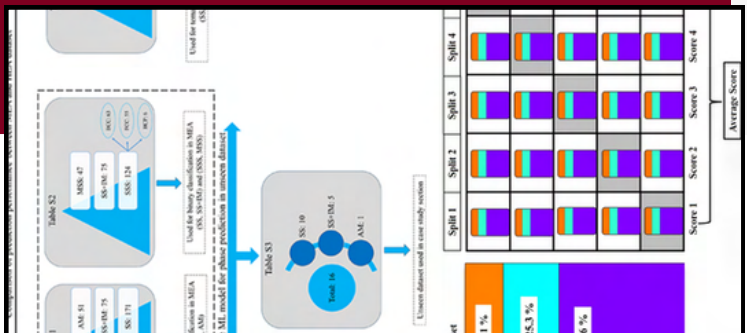
Micro and Nano Engineering



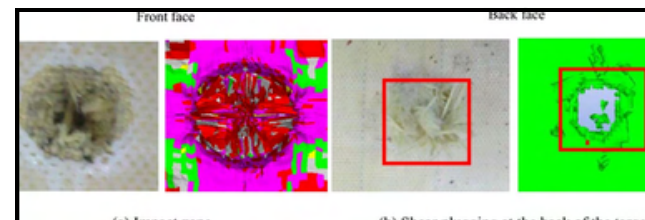
In-situ monitoring of molten pool shape during remote laser beam welding



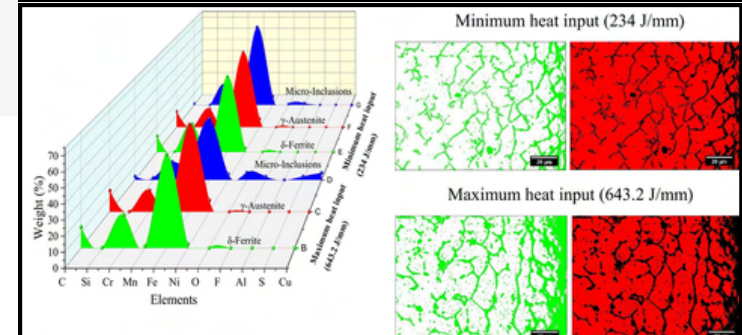
Advanced Tool Development



Machine learning based phase prediction of medium entropy complex alloys



Composite Design and Modelling



Evolution of δ -ferrite grain morphology of SS308L fabricated using WAAM-CMT

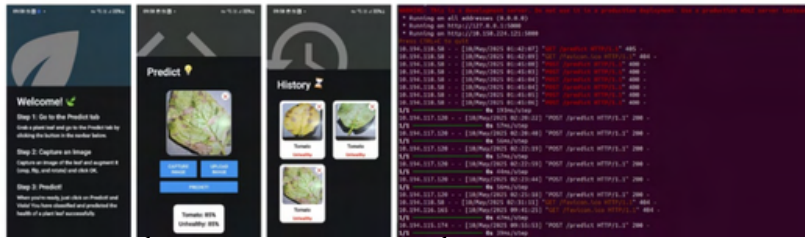
Production Engineering

Glimpses of our Projects



- Implementing Digital twin of a Machine Tool.
- Predictive Quality Analytics for turn component using IIoT & Machine Learning.
- Composite 3D Printing by VAT Photopolymerization.
- Design and fabrication of metamaterial based lightweight acoustic panels.
- Energy storage system fabrication using second life battery.
- Metal additive manufacturing for EV application.
- Development of lightweight ceramics for high energy defeat off by sniper projectiles.
- Machining of special grade materials for Automobile applications.
- Experimental investigation and numerical modelling of laser beam welded aluminum joints for EV Batteries.
- Design, Fabrication and experimental investigation of lattice based metal foams using additive manufacturing technique.
- Experimental investigation and numerical modelling of laser beam welded aluminium joints for space-grade batteries
- Design, Fabrication and experimental investigation of lattice based metal foams using additive manufacturing technique
- Design and fabrication of metamaterial based lightweight acoustic panels
- Topology optimisation of lattice materials for Additive manufacturing
- Development of lightweight ceramics for high energy defeat off by sniper projectiles.

DEPLOYMENT



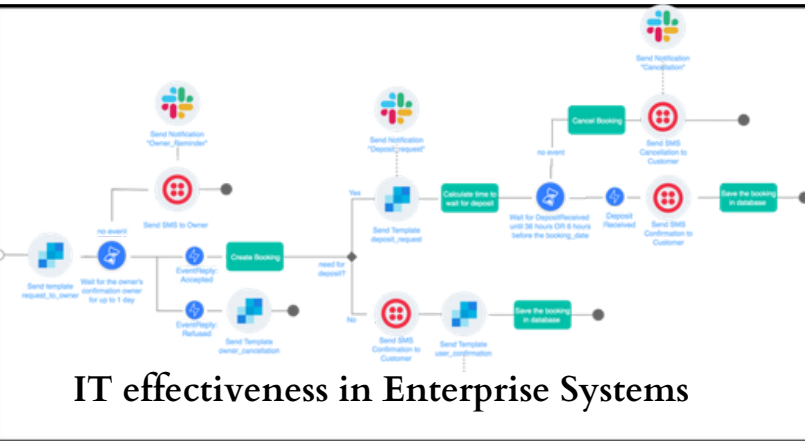
Machine Learning and Deep Learning



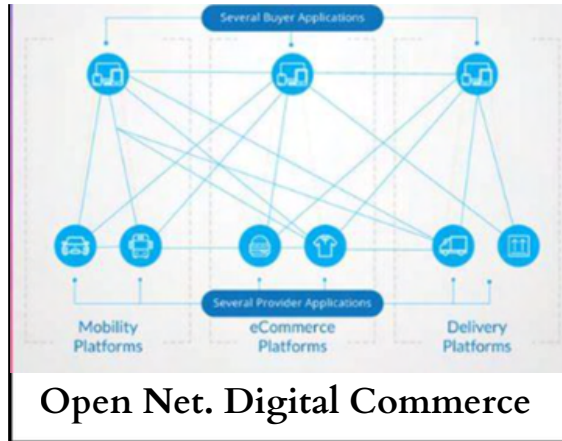
Intelligent Manufacturing

Glimpses of our work / Research

Industrial Engineering



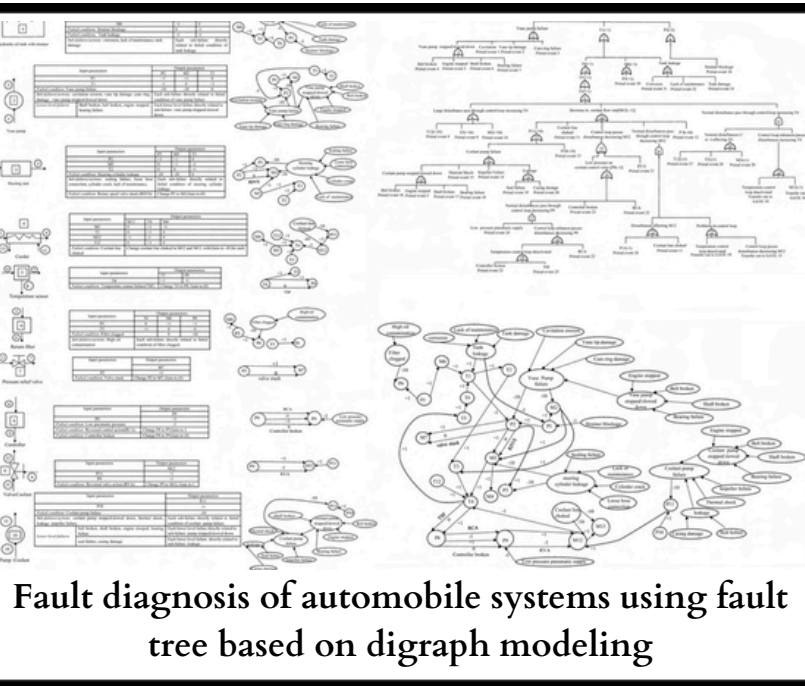
IT effectiveness in Enterprise Systems



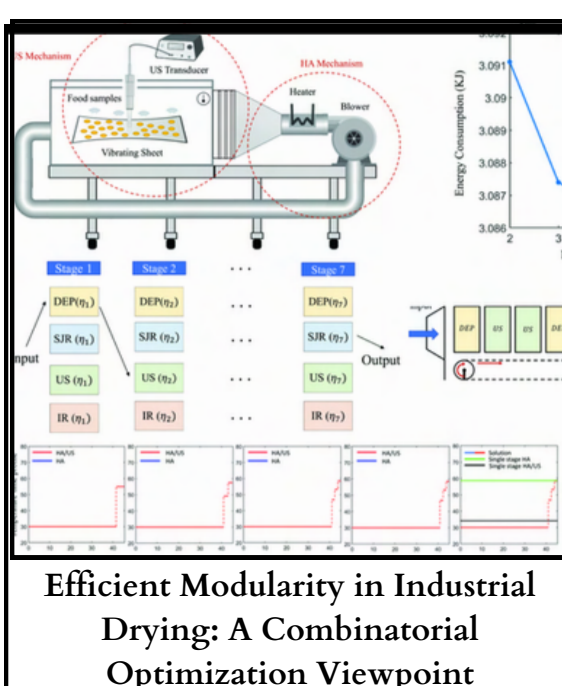
Open Net. Digital Commerce



Operation Research



Fault diagnosis of automobile systems using fault tree based on digraph modeling



Efficient Modularity in Industrial Drying: A Combinatorial Optimization Viewpoint



Optimization

Industrial Engineering

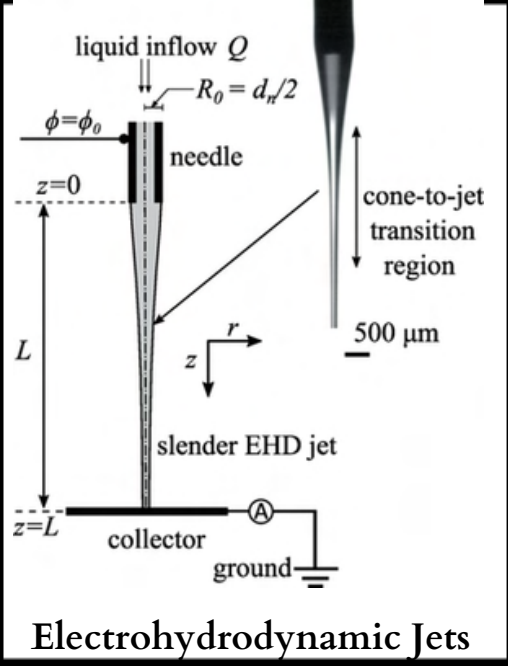
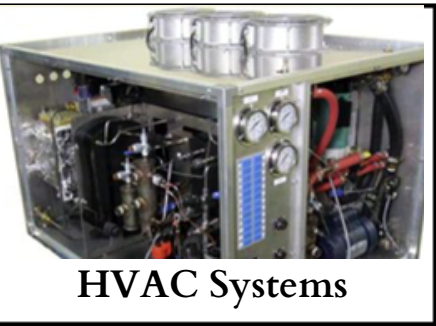
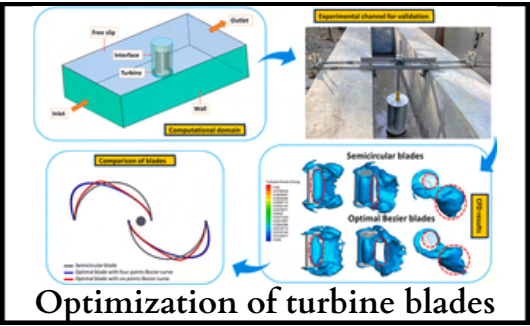
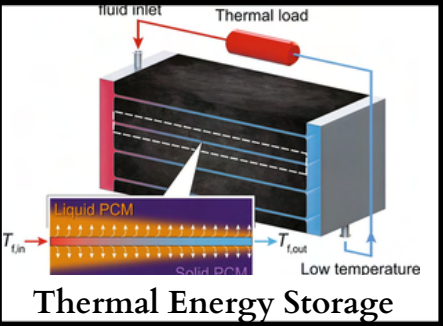
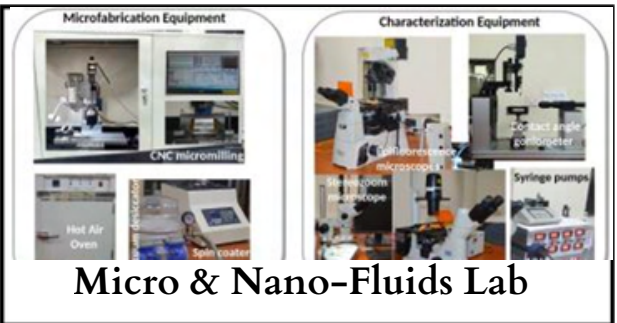
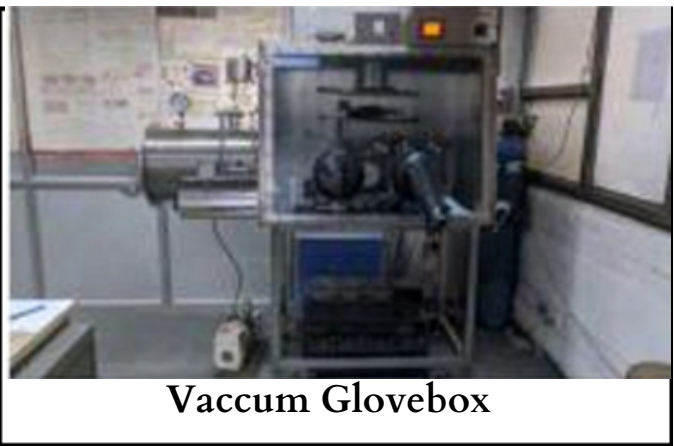
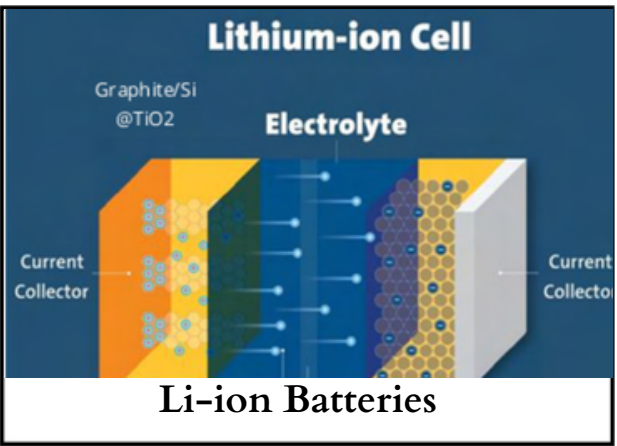
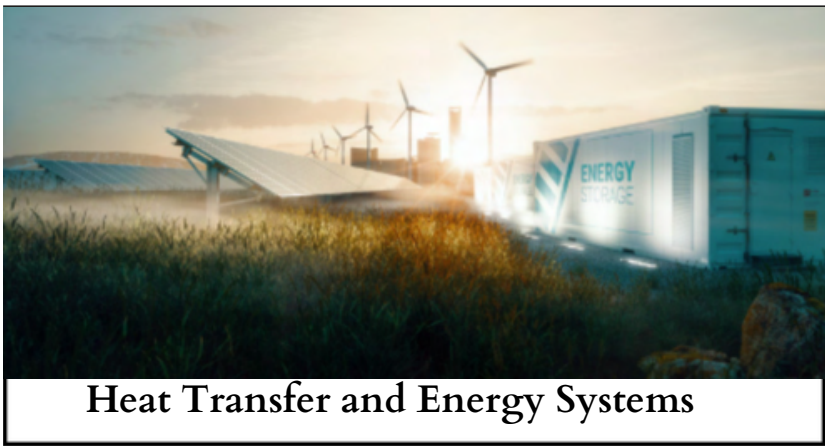
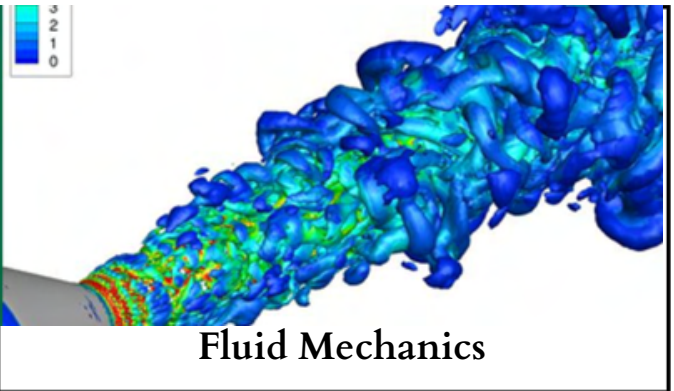
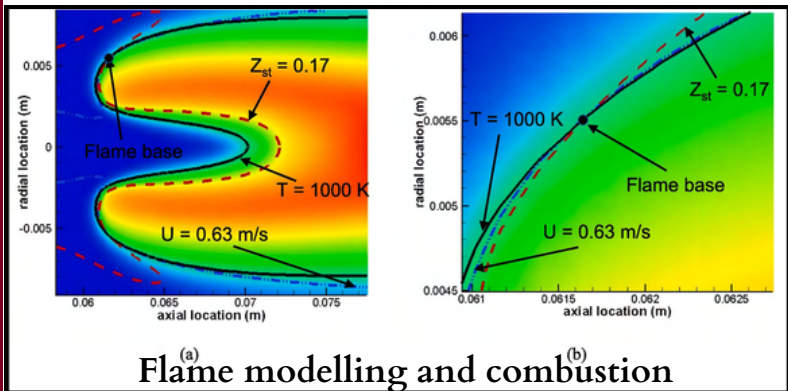
Glimpses of our Projects



- Development of IIoT systems for connected CNC machines and performance analytics.
- Digital Twin for Process Industry.
- Deep Learning based Video Analytics for Manufacturing Applications. Predictive Model for Estimating the Remaining useful lifespan of Machines Using Machine Learning.
- Develop an assessment model to evaluate the readiness level of Indian MSMEs for Industry 4.0 adoption.
- Electric Bus Business Model Optimization.
- Implementing Autoware for Autonomous driving
- Packing Optimization.
- Precision Timetabling For Indian Railways
- Logistics Management
- AI/ML Models for Lung Cancer Detection Using CT Scan Data.
- Integer valued portfolio optimization incorporating machine learning inputs.
- Machining Learning model for MRI registration.
- Driving success in tennis through data analytics.
- Forecasting the Future Market Dynamics of ONDC in India.

Glimpses of our facilities / Research

Thermofluids Engineering





Thermofluids Engineering

Glimpses of our Projects

- Design of Hybrid Electric Vehicle layout for Delhi NCR urban vehicle traffic. HVAC Front and Rear Cooling/Heating System Analysis.
- Design of inlet and isolator of dual mode scramjet engine through CFD. Investigation of the effusing cooling of the combustor liner aero engines. Using image processing for interfacial profiles and inferences in interfacial fluid mechanics.
- Computational Fluid Dynamics Simulations for Immersed Body Problems. Drag reduction system(DRS) for FS race car(rear wing).
- Simulation of Li-plating in a Lithium-ion cell using Pybamm.
- Design and testing of an industrial burner for hydrogen fuel.
- Optimizing Ethanol-Petrol Ratios for Improved Urban Traffic Fuel Economy and Environmental Impact.
- Experimental testing of a V-gutter based flame stabilizer for aircraft afterburner applications.
- Soot modelling of sustainable aviation fuels.
- Convention cooling of the high power PCBs arrays
- CFD simulation of Heat transfer in laminar to turbulent transition in free and mixed convection Regimes.
- Microfluid devices for electro-chemical energy conversion.



B.Tech Programs

There are two undergraduate degree programs offered by the department.

Students begin with institute core courses in basic sciences — Physics, Chemistry, Mathematics — then move to engineering mechanics, computer science, environmental science, electrical engineering, materials science, numerical methods, statistics and humanities.

Rigorous laboratory classes in 3rd and 4th year promote hands-on learning. In the final year, students execute a semester-long B.Tech. project in direct collaboration with faculty research and industrial design work.

Mechanical Engineering Courses

- Solid Mechanics
- Fluid Mechanics
- Kinematics & Dynamics of Machines
- Manufacturing Processes
- Engineering Thermodynamics
- Design of Machines
- Control Theory and Applications
- Energy Systems and Technologies
- Heat and Mass Transfer
- Introduction to Operations Research
- CAM and Automation
- CAD and Finite Element Analysis
- Manufacturing System Design
- Introduction to Statistics

Tools & Software

ANSYS

Hypermesh

Autodesk Inventor

SolidWorks

MATLAB

Python

C/C++

SQL

MongoDB

Excel

Tableau/PowerBI

SPSS

CPLEX/Gurobi

AnyLogic

Google OR Tools

Production & Industrial Engg. Courses

- Solid Mechanics
- Kinematics & Dynamics of Machines
- Metal Forming and Press Tools
- Near Net Shape Manufacturing
- Metrology and Quality Assurance
- Welding and Allied Processes
- Material Removal Processes
- Thermal Science for Manufacturing
- Stochastic Modelling and Simulation
- CAM and Automation
- Micro and Nano Manufacturing
- Composite Materials & Manufacturing



Structural and Mechanical Design

- Energy absorption capabilities of gyroid structures
- Open cell structures for high fatigue strength and low weight Design and development of a Smart Beam
- Design of cable car cabin
- Design of 3D Food Printer
- Table top fatigue testing machine for miniature samples Damping in blisk design using split rings
- Modular wind turbine design and fabrication
- Design of Sheep Shearing Device and Carpet Looms
- Compact cost-effective lift for 2-story village houses

Biomedical Engineering

- Biomechanics of total shoulder replacement
- Tibial keel design and knee replacement – FEA study
- Variable size detachable cannula for surgery under microscope Wave propagation in bone structure

Control Systems and Automation

- Control of structural systems using reinforcement learning
- Control of a Flying Unicopter with Payload Variation
- Imitation Learning from MPC for Quadrupedal Multi-Gait Control
- Predictive Maintenance using AI/ML

Traffic and Simulation

- City-wide simulation of traffic dynamics for heterogeneous fleet
- Simulation modeling, control, and planning for non-Newtonian fluid robots Metamodel based network simulation

Logistics and Supply Chain

- ONDC based Logistics Business Model
- Re-Commerce Supply Chain Optimization Model
- AI integration with Supply Chain and Optimization
- Applications of Learning and optimization in logistics

Digital and Smart Manufacturing

- Dimension measurement of object using computer vision
- Track check in Helicopter Main Rotor Blade using image processing

Optimization and Mathematical Programming

- Capacitated Simultaneous Transportation & Production Problem
- Workload balancing via mathematical programming
- Generalized Nash Equilibrium Problems with Mixed-Integer
- Variables Constraint programming for railway scheduling
- Multiobjective optimization for concentrating solar power plant
- Parallel Algorithms for Mixed-Integer Nonlinear Optimization
- Portfolio Optimization
- Topology optimization using integer and nonlinear programming

AI, Machine Learning & Data Processing

- Advanced Applications of AI-based Data Processing
- Neuroscience-inspired Graph Neural Network for computational mechanics

M.Tech. Mechanical Design

WHY US?

Mechanical Design encompasses creating, analysing, and refining products from simple parts to complex machinery — emphasising high performance, manufacturability, and better testing criteria. Students acquire skills through industry R&D projects, focusing on problem-solving, critical thinking, and hands-on experience with CAE tools, prototyping, testing, and system validation.

Areas of Expertise

Finite Element Analysis	Automotive Crash Safety	Impact & High Strain Rate	NVH
Vibration & Noise Control	Condition Monitoring	Rotor Dynamics	Tribological System Design
Optimized Bearing Design	ML Implementation	Robotics and Control	Advanced Mechanisms
Biomedical Engineering	Human Body Modeling	Impact Biomechanics	

Key Projects

- Biomechanical evaluation of implant failure; high strain-rate crash & fracture analysis
- Car model for crash & durability applications; fracture analysis of semiconductor devices
- Impact and blast attenuation of cellular structures
- Ground Satellite Handling System with two-axis C.G. adjustment
- Noise & vibration analysis of balancing piston of aeroengine rotor
- Investigating noise and design of low noise tyres
- Compliant-based robotic hand for grasping; micro-actuation

Courses

- Optimum Design
- Mechanical System Design
- Dynamics and Control
- Finite Element Method
- Automotive structures Design
- Machine Learning for Computational Design
- Rotordynamics
- Robotics
- Acoustics and Noise control
- Orthopedics and Implant Design
- Experimental Modal Analysis



Tools

- ANSYS
- ABAQUS
- LS-DYNA
- COMSOL Multiphysics
- Hypermesh
- MATLAB
- Rhino
- Radioss
- MIMICS
- RecurDyn

M.Tech. Production Engineering



WHY US?

The curriculum includes traditional subjects enhanced by advanced computational tools to optimise the process. Students learn modern manufacturing techniques through projects and hands-on work on advanced machines. Interdisciplinary projects (biomedical devices, industrial networking, etc.) enable a multitude of subjects.

Areas of Expertise

Scientific Computing / FEA / Optimisation

IIoT

ML in Manufacturing

Cyber-Physical Systems

MES

Additive Manufacturing

Wire Arc Additive Manufacturing

Metal Joining & Forming

Surface Engineering

Composite Materials

Polymeric Stents

Key Projects

- Online wear monitoring and Tool Failure Prediction using ML and Simulation Welding quality
- analytics using Deep Learning
- High Entropy Alloys – Nanoparticle fabrication & characterisation
- Development of Hard coatings using PVD
- Machining of aerospace materials using coated tools; Friction Stir Processing
- Post heat treatment study and characterisation of WAAM single wall structures
- Additive Manufacturing of advanced cutting tools; 3D Printing of Supercapacitors
- Design and fabrication of Auxetic sport protectives by Additive Manufacturing

Courses

- Machining Process and Analysis
- Welding Science Technology
- Metal Forming
- Computer Aided Manufacturing
- Computational Methods
- Experimental Methods
- Additive and Smart Manufacturing
- Processing & Mechanics of Composites
- Industrial Engineering Systems
- Supply Chain Management

Tools

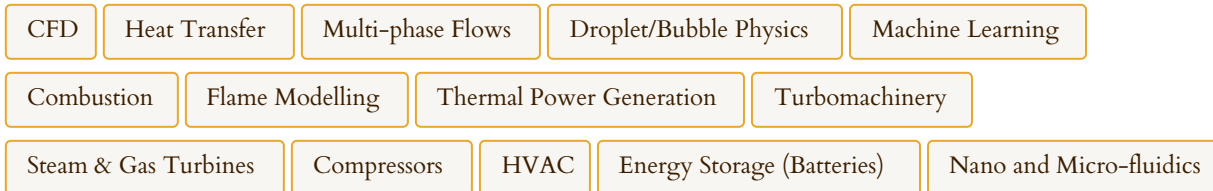
- ANSYS / ABAQUS / LS-Dyna
- SolidWorks / Autodesk Fusion
- MATLAB / Python
- ThingWorx / KepWare / UA Server
- RoboDK / ROS / Unity
- LabView

M.Tech. Thermal Engineering

WHY US?

Advanced training in heat transfer, thermodynamics, fluid mechanics, energy systems, combustion, and micro-fluidics. A blend of theoretical knowledge with practical applications through state-of-the-art labs and computational facilities. Research supported by industry and government partnerships prepares students for aerospace, automotive, energy, and manufacturing sectors.

Areas of Expertise



Key Projects

- Two-phase flow solver with adaptive mesh; IVP solvers for transient multiphase flows Turbulent kinetic energy during shock-driven combustion of droplet clusters
- Droplet Splashing Over a Moving Porous Substrate; EHD instabilities
- Propeller profile optimization for noise reduction in UAVs
- Co-firing thermal power plants with biomass; spray flame interactions
- Specific Gasification Rates of biomass fuels for TLUD gasifier
- Thermochemical Energy Storage System for HVAC
- Feasibility of central district cooling/heating for IIT Delhi campus

Courses

- Computational Heat Transfer
- Combustion
- Gas Dynamics
- HVAC
- Thermal Design
- Radiation Heat Transfer
- Convective Heat Transfer
- Turbo Compressors
- Heat Exchangers
- Steam and Gas Turbines
- Advanced Fluid Mechanics
- Advanced Thermodynamics
- Design of Wind Power Farms
- Aerodynamics of Flight
- Physics of Turbulent Flows
- Renewable Energy



Tools

- ANSYS Fluent
- OpenFOAM
- COMSOL
- ICEM CFD
- PTC Creo
- SolidWorks
- LabVIEW
- EES

• MATLAB / Python / C/C++

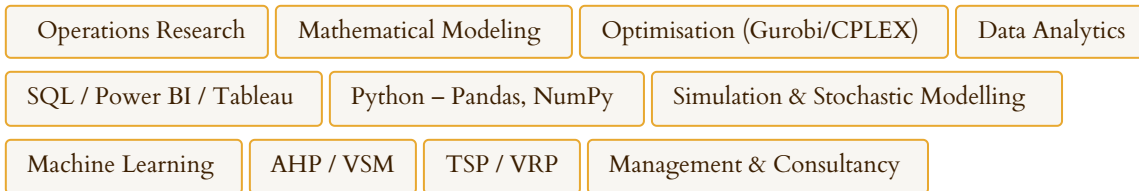
M.Tech. Industrial Engineering



WHY US?

The course offers a balanced combination of theoretical rigour and practical hands-on experience with modern solvers, software, and projects. Students receive comprehensive training in industrial engineering, optimisation, and operations research to address complex, contemporary techno-managerial challenges.

Areas of Expertise



Key Projects

- Data-driven analysis of Indian Judicial System using ML and Mathematical Modelling
- Simulation and dynamic scheduling of APM-type operations
- Capacity planning for cancer care provision in India
- Pharmacy store transaction analytics for inventory management and disease profiling
- Deep Learning for lung cancer detection
- Optimisation of India's public distribution system using VRP and MILP
- Network design for sustainable future / entropy-based methods
- Simulation and optimisation of hospital appointment scheduling
- Rake linking optimisation; Precision Timetabling for Indian Railways

Courses

- Operations Research (OR)
- Advanced OR
- Stochastic Modeling & Simulation
- Probability & Statistics
- Supply Chain Management
- Industrial Engineering Systems
- Machine Learning
- Operations Planning and Control
- Project Management

Tools

- Python / R / C++
- SQL / MongoDB
- Excel / Tableau / PowerBI
- SPSS
- CPLEX / Gurobi
- AnyLogic / MATLAB
- Google OR Tools



M.S.(R) Mechanical Engineering

The M.S.(R) is a master's-level program with a more research-centric approach than M.Tech. Students enjoy greater flexibility in course credit requirements and devote most of their time to active research.

All students graduate with a degree of **M.S.(R) in Mechanical Engineering**.

● About the Program

Students do their projects primarily in four specialisations — Thermal, Design, Production, and Industrial Engineering — and may also engage in inter-disciplinary research based on their interest.

● Key Projects

- Mechanical Design:** Online wear monitoring and Tool Failure Prediction using Machine Learning and Simulation
- Thermal Engineering:** High Entropy Alloys – Nanoparticle fabrication and characterisation study
- Production:** Ultrasonic Joining of EV Batteries
- Industrial Engineering:** ONDC (Open Network for Digital Commerce) for e-commerce and logistics in India

● Courses & Tools

Same as individual M.Tech programs. Please refer to the M.Tech Mechanical Design, Thermal, Production, and Industrial Engineering sections for full course and tool details.



V.L.F.M Mechanical Engineering

Programme Vision

To create a distinct set of visionary leaders with the ability to contextualise and contribute to the nation's manufacturing renaissance.

PGDEx-VLFM is a one-year full-time residential executive diploma programme jointly conducted by **IIM Mumbai** and **IIT Delhi** to groom emerging leaders for mid and senior-level managerial positions.

Scope of the Programme: The programme thrives on peer-learning pedagogy focusing on management, manufacturing technology and leadership. The curriculum is built on Breakthrough Management and the Five-Step Discovery process, enabling participants to shift from "small m to big M" and to look at the invisible through the Third Eye.

National and global economies face challenges on a scale unseen in nearly a century. Post-pandemic, the world moves towards a new era of manufacturing renaissance. The nation aims to increase the manufacturing sector's contribution to GDP at a growth rate of **25% by 2025**, where the new generation of leaders with vision will play a pivotal role.

The programme significantly contributes to the **Make in India** and **Atmanirbhar Bharat** initiatives by nurturing strategic, techno-managerial talent — including **1,300+ contact hours**, industry internships, and global exposure.



Management & Strategy

- Manufacturing Strategy
 - Operations Research in Manufacturing
 - Strategic Planning and Implementation
 - Cost Management and Controls
 - Managerial Economics
 - Data Analytics
 - Marketing: Theory and Practice
 - Advanced Human Resource Management
 - Sustainable Strategies for Manufacturing
 - Market Research for Global Manufacturing
- Competitiveness

Technology & Operations

- Core Manufacturing Processes
 - Statistics for Manufacturing Management
 - Artificial Intelligence and Machine Learning
 - Additive Manufacturing
 - Automation and Smart Manufacturing
 - Energy Management and Industrial Safety
 - Supply Chain, Strategic Sourcing, and Logistics Planning
- Strategic Imperatives of New Product Development



Ph.D. Mechanical Engineering

The Ph.D. involves a deep understanding of theoretical concepts and applying them to solve complex problems aligned with the Sustainable Development Goals (SDG).

Three groups drive this mission:

Design Group · ThermoFluid Group · Production & Industrial Group

Students acquire deep understanding of fundamental technical knowledge, planning and managing resources for conducting industry R&D projects.

<i>Design Group</i>	<i>Thermofluid Group</i>	<i>Production & Industrial Group</i>
• Finite Element Analysis • Noise Vibration & Harshness • Vibration and Noise Control • Condition Monitoring • Rotor Dynamics • Digital Twin of machineries • Biomedical device & implant design • Flexible Manipulators • Training module for surgery	• CFD & Heat Transfer • Multi-phase flows • Droplet/Bubble Physics • HVAC • Pollution mitigation strategies • Energy Storage (Solar/Batteries) • Nano and Micro-fluidics	• Bio-Materials for Implants & Ortho. • Additive Manufacturing / 3D-4D Printing • Sintering • Blast Mitigation Material Systems • Sustainable Supply Chains

All courses from M.Tech. Design, Thermal, Production and Industrial programs are available, along with other department courses. Students gain deep expertise in FEA and other computing software.



Design Group

Noise Vibration & Harshness

- Condition monitoring of wind turbine power train
- Noise & vibration analysis of Epicyclic gearbox
- Digital twin for rotor and gearbox systems

Biomedical Device

- Dynamics of Highly Flexible Manipulators
- Development of a heterogeneous lung model
- for blunt impact loads
- Flexible manipulator for Neurosurgery

Training module for prostate needle biopsy

Thermofluid Group

CFD / Heat Transfer

- Induced flow patterns in flapping wing aerodynamics
- Fluid flow over patterned and lubricated surface textures
- CFD simulation of micro-heat exchanger

Solar Energy & Storage

- Porous Volumetric Solar Receiver Parabolic
- Solar Collector
- Thermal characterisation of commercial lithium-ion batteries
- Composite Anode for Li-ion Batteries (Graphite/Si@TiO₂)
- Simulation of Solar Absorption Cooling System

HVAC & Pollution

- V-Bar splash fills-in wet cooling towers performance analysis
- Mitigation of air pollution by uncharged and charged droplets

Production & Industrial Group

Additive Manufacturing

- Novel entropy alloys for implants
- 4D-printed shape memory polymer composite – biomedical
- Fabrication of Magnesium alloys for Implants
- Fabrication of Biodegradable Composite Implants

Blast Mitigation

- Material Characterisations under Shock Wave Loading
- High Strain Rate Material Test Setup
- Blast Mitigation properties of materials

Sustainable Energy

- Reverse Logistics Modelling for Electric Vehicle Batteries

Thermal Sciences & Energy Systems

- CFD simulation of effusion cooling of gas turbine combustor
- Experimental investigation of nucleate boiling over enhanced surfaces
- CFD study on heat transfer enhancement in parabolic trough receiver
- Modeling and analysis of thermal storage integrated supercritical CO₂ refrigeration systems
- Studies on immersion cooling of Lithium-ion cells
- Computational Analysis of a Wing Surface Heat Exchanger
- Analysis of Pillow Plate Heat Exchangers with aerofoil welds
- Electromagnetic control of heat transfer in high-speed steel billet continuous casting
- Co-firing thermal power plants with biomass for cleaner power

Combustion & Propulsion

- CFD analysis of unstart in scramjet inlet
- Shock-induced ignition of heterogeneous hydrogen and air mixture
- Turbulence production in shock-particle interaction
- Experimental & numerical study of bluff body combustor for aircraft afterburner
- Modeling soot formation from lubrication oil in hydrogen engines
- Flame temperature measurement using cost-effective color ratio pyrometry
- Reactor network modeling of biomass-based kilns and furnaces

Design & Advanced Manufacturing

- Process simulation and tooling design for toroid forming (aerospace)
- Wire arc additive manufacturing (WAAM) for metal-based composites
- Composite 3D Printing by VAT Photopolymerization
- Ultrasonic-assisted composite 3D Printing
- Design & fabrication of lattice-based metal foams via additive manufacturing
- Design & fabrication of metamaterial-based lightweight acoustic panels
- Topology optimisation of lattice materials for additive manufacturing
- Polygonal machining for medical implants
- Substrate-supported nanoparticles through dewetting methods
- Diffusion bonding of similar and dissimilar materials

Biomedical & Fluid Systems

- Biological micro-flows: pulmonary and portal capillary beds
- Modelling bubble dynamics in tissues (histotripsy, drug delivery)
- Modeling inhalation aerosol deposition in lungs
- Design and performance of an indoor air purifier
- Data-driven modelling of drone dynamics (quadcopter)

Industrial Engineering & Analytics

- Data-driven analysis of Indian judicial system using ML & mathematical modeling
- Simulation and dynamic scheduling of APM-type operations
- Capacity planning for cancer care provision in India
- Pharmacy store transaction data analytics for inventory & disease profiling
- Deep learning in healthcare — lung cancer detection
- Optimization of India's public distribution system (VRP & ILP/MILP)
- Network design for sustainable future using entropy-based methods
- Simulation & optimization of appointment scheduling in hospitals
- Rake linking optimization
- Developing hedging strategies for incomplete markets with RL

Joining & Surface Engineering

- Advanced joining of special-grade materials for defense applications
- Machining of special-grade materials for automobile applications
- Development of novel coating for cutting tool applications
- Investigating surface texturing on weldability of metal-to-polymer (laser)
- Laser beam welded aluminium joints for space-grade batteries
- Study of advanced metal transfer modes during WAAM

Energy, Environment & Renewables

- Solar radiation modeling & prediction of renewable energy potential
- CFD modeling of axial flow compressors (aircraft & stationary gas turbines)
- Modeling & analysis of thermal storage integrated CO₂ refrigeration systems
- Co-firing thermal power plants with biomass for cleaner power

Defence & Structural Applications

- Development of high energy projectiles
- Development of lightweight ceramics for high-energy defeat of sniper projectiles
- Bluff body combustion for military aircraft engine afterburner
- CFD analysis of unstart in scramjet inlet

Advanced Materials & Nanoscience

- Substrate-supported nanoparticles via dewetting methods
- Design, fabrication and experimental investigation of lattice-based metal foams
- Metamaterial-based lightweight acoustic panels
- Topology optimisation of lattice materials for additive manufacturing

Past Recruiters



For further information: <https://ocs.iitd.ac.in/ocs/recruiter.php?id=past>



Placement Procedure

1

Contact Faculty / Placement Officer

Interested companies contact the Faculty Coordinator or OCS for a Job Notification Form (JNF) at placement@admin.iitd.ac.in

2

JNF Received

Once the filled JNF with all required details is received, companies are assigned a username/password to access their online account on the OCS website.

3

IIT Delhi Sends an Invite

Once you show interest, an email is sent by the placement office with relevant details and a registration link.

4

Companies Fill the JNF / TNF

JNF (Job Notification Form) and TNF (Training Notification Form) needs to be filled and frozen. Opening is then listed in the IITD Recruitment Portal.

5

Company Uploads Content

Companies are assigned server space to upload presentations, videos, and any information they want students to see.

6

Students Apply

After the JNF is frozen, eligible students can view details and apply. Resumes are visible to the company after the application deadline.

7

Company Submits Shortlist

Company submits a shortlist on its online account before a deadline. Shortlisted students get notified.

8

Shortlist, Interview & Recruit

The placement office allots campus interview dates. Students apply via the portal; recruiters shortlist, interview, and hire.

Contact us



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